

Stephen Zhang

syz@stanford.edu | Palo Alto, CA

EDUCATION

Stanford University <i>Bachelor of Science in Mathematics and Computer Science</i> Minor: Music (Trumpet Performance) Relevant Coursework: Partial and Stochastic Differential Equations, Artificial Intelligence, Statistical and Machine Learning, Algorithm Design, Real Analysis, Probability Theory, Discrete Mathematics	Stanford, CA <i>Graduation Date – June 2029</i>
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EXPERIENCE

Alpha AI Engineering <i>AI Engineering Intern</i> - Built full-stack web applications in TypeScript, React, and Firebase, using Firestore, Cloud Functions, object storage, and federated authentication - Modified the Rust scheduling backend of an Anki fork, adding a SvelteKit interface, a self-hosted sync server, and an Android build - Fine-tuned a sub-1B-parameter language model with LoRA, using staged supervised fine-tuning and direct preference optimization, and graded its output by execution rather than by model scoring - Ran transformer pretraining sweeps on a SLURM cluster in PyTorch, covering corpus generation, checkpointing, and evaluation harnesses	Austin, TX <i>June 2026 – August 2026</i>
EquityBound <i>Intern</i> - Analyze small and medium enterprise valuations using standard financial methods - Propose and implement service enhancements with the engineering team - Audit the codebase and identify accuracy and performance improvements	Remote <i>January 2026 – Present</i>

RESEARCH EXPERIENCE

Feldman Lab, Department of Biology, Stanford University <i>Research Intern</i> - Develop and analyze stochastic, agent-based, and equation-based models of gene–culture coevolution using Python (NumPy, SciPy, Matplotlib) - Derive and validate analytical results against simulations - Published one peer-reviewed paper, with an additional manuscript under review	Stanford, CA <i>June 2023 – Present</i>
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PUBLICATIONS

- Laurel Fogarty, Stephen Zhang, Marcus W. Feldman (2025). Gene-culture association and coevolution. Theoretical Population Biology, Vol. 165, pp. 62–71 - Smarties Sandwiches (2024). Pi in the Sky, Pacific Institute for the Mathematical Sciences (PIMS), issue 23, pp. 24–27	
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SKILLS

- Programming: Python, C++, Java, TypeScript, R, SQL, LaTeX - Frameworks and tools: PyTorch, Hugging Face, React, Firebase, AWS, SLURM, NumPy, SciPy - Languages: English (native) and Mandarin Chinese (fluent reading and writing)	
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EXTRACURRICULAR ACTIVITIES

Stanford Orchestras <i>Financial Officer</i> - Oversee orchestra activities as a member of the orchestra executive committee with the co-presidents and vice president - Manage a budget of more than \$40K, ensuring all events have funding and limiting excess spending	Stanford, CA <i>September 2025 – Present</i>
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AWARDS

- Stanford Math Tournament – Distinguished Honorable Mention (Top 10%) - USA Physics Olympiad Semifinalist (Top 400 in U.S.) 5x AIME Qualifier - Fellow of the Royal Schools of Music in Trumpet Performance - CLASS National Chinese Essay Contest Gold Prize	2024 2022 2020 – 2025 2025 2023
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